

Project Management Tool

(Trello Clone)

SDE Intern Fullstack Assignment

Description

Build a Kanban-style project management web application that closely replicates Trello's design and user experience. The application should allow users to create boards with lists and cards, organize tasks using drag-and-drop, and manage project workflows visually. Your implementation should resemble Trello's UI patterns, layout structure, and interaction design.

AI Tools Usage

You are **allowed and encouraged** to use AI tools such as ChatGPT, Claude, GitHub Copilot, Cursor, or any other AI assistants for development. However, you **must understand every line of code** you submit and be prepared to explain your implementation decisions during the evaluation interview.

Technical Stack

- **Frontend:** React.js or Next.js (Single Page Application)
- **Backend:** Node.js with Express.js OR Python with FastAPI/Django
- **Database:** PostgreSQL or MySQL (design your own schema)

Core Features (Must Have)

1. Board Management

- Create a board with a title
- View board with all its lists and cards

2. Lists Management

- Create, edit, and delete lists
- Drag and drop to reorder lists

3. Cards Management

- Create cards with a title
- Edit card title and description
- Delete or archive cards
- Drag and drop cards between lists
- Drag and drop to reorder cards within a list

4. Card Details

- Add and remove labels (colored tags)
- Set due date on cards
- Add checklist with items (mark items as complete/incomplete)
- Assign members to cards

5. Search & Filter

- Search cards by title
- Filter cards by labels, members, or due date

Good to Have (Bonus)

- Responsive design (mobile, tablet, desktop)
- Multiple boards support
- File attachments on cards
- Comments and activity log on cards
- Card covers (images)
- Board background customization

Important Notes

- **UI Design:** Your application should closely resemble Trello's design. Study Trello's UI carefully before starting - visit trello.com to understand the user flows and interface patterns.
- **Drag & Drop:** The drag and drop functionality is a core feature. Ensure smooth interactions for reordering lists and moving cards.
- **No Login Required:** Assume a default user is logged in. Create sample members in the database for assignment functionality.
- **Sample Data:** Seed your database with a sample board, lists, cards, and members.
- **Database Design:** Design your own database schema. This will be evaluated.
- **README File:** Include setup instructions, tech stack used, and any assumptions made.
- **Original Work:** Plagiarism from existing repositories will result in immediate disqualification.

Submission

- Upload your code to GitHub and ensure the repository is **public**
- Deploy your application (Vercel, Netlify, Render, Railway, or any cloud service)
- Submit both the GitHub repository link and the deployed application link

Evaluation Criteria

Criteria	What We Look For
Functionality	All core features working correctly
UI/UX	Visual similarity to Trello's design and UX patterns
Drag & Drop	Smooth and intuitive drag-and-drop interactions
Database Design	Well-structured schema with proper relationships
Code Quality	Clean, readable, and well-organized code
Code Modularity	Proper separation of concerns, reusable components
Code Understanding	Ability to explain your code during evaluation

Timeline

Submission Deadline: 2 days from the date of receiving this assignment

Good luck!